A Smart-Contract Based Protocol for Transferable Hedging of Concentrated Liquidity Positions

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Abstract

Decentralized Exchanges are a core component of Decentralized Finance. Liquidity providers play a key role in ensuring that a given token pair is liquid. However, liquidity provision exposes participants to impermanent loss and to the depreciation of liquidity positions caused by adverse price movements. This problem is particularly severe in concentrated liquidity market makers, where impermanent loss impact is more extreme for tighter price ranges. Existing approaches address this issue through portfolio optimization or derivatives-based hedging strategies, but they usually depend on liquid derivatives markets and rely on approximation to hedge the position across the equity line. They do not offer a protocol-level mechanism for direct risk transfer. This is a crucial limitation, because derivatives like options are not available for several token pairs, and perpetuals or futures offer linear payouts, which are not suitable for hedging liquidity positions. This paper addresses that gap by proposing a smart-contract-based protocol that enables liquidity providers to transfer liquidity position depreciation risk to counterparties in exchange for a payoff. Following a Design Science Research methodology, we develop a taxonomy of existing risk mitigation and hedging approaches, a goal model of the protocol, a dynamic model of the system and a proof-of-concept prototype of the risk transfer protocol. We then evaluate the artifact by means of use case demonstration and benchmark the profit and loss with existing appraoches.

Keywords

DeFi, Automated Market Makers, Concentrated Liquidity, Impermanent Loss

1. Introduction

Automated Market Makers (AMMs) have become a foundational element of decentralized finance (DeFi), enabling permissionless exchange without centralized intermediaries [1]. In these systems, liquidity providers (LPs) deposit assets into liquidity pools and earn fees from trading activity. While this model has unlocked broad participation in market making, it also exposes LPs to substantial financial risk. In particular, LPs have non-linear returns, due to the impermanent loss when asset prices move away from the initial deposit ratio [1]. In concentrated liquidity market makers (CLMMs), this risk becomes even more pronounced because the value depreciation of a position also depends on the range length of choice [2].

The state of the art offers several contributions addressing the need to effectively mitigate risk exposure for LPs. Prior work studies mathematical representations of LP payoff functions [2], dynamic rebalancing strategies [3], hedging strategies using options [4], and portfolio optimization methods [5]. However, these approaches show two important shortcomings. First, many of them rely on external derivatives markets, which are unavailable or illiquid for a large number of token pairs [4]. Second, dynamic replication strategies face practical constraints, due to the need of frequent inventory adjustments, incurring in trading fees [3]. Ultimately, the proposed approaches support position management at the user level rather than implementing a protocol tailored for liquidity provision that

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allows direct transfer of liquidity-position risk. This paper addresses this gap by responding to the following Research Question (RQ): *How can a smart-contract protocol be designed to enable transferable hedging of the equity of CLMM liquidity positions?* To address this RQ, the paper makes three main contributions. First, it develops a taxonomy of existing strategies to hedge liquidity positions. Second, based on the taxonomy identified, it models the requirements and the architectural design of a smart contract based protocol that allows users to transfer the risk of liquidity positions in exchange for a premium. Finally, we design a suitable pricing mechanism for pricing the risk premium.

2. Methodology

We follow a Design Science Research (DSR) methodology in which the goal is to create and evaluate a set of technological artifacts [6]. The first contribution systematizes the current state of knowledge and highlights the unresolved design space [7]. Based on this taxonomy, we elicit the requirements of the protocol and model them using the Trusted DApp Modeling approach[8], an extension of goal model diagrams that facilitates the specification of on-chain components of the system and their mapping to compoent diagrams. Based on this model, we also derive the component diagram of the system. Third, we use Coloured Petri Nets (CPN) [9] to specify the dynamic process of risk transfer, and specify a pricing mechanism that mactches dynamically demand and supply and reacts to relevant risk drivers.

Based on the DSR methodology, the designed artifact is evaluated by implementing a prototype based on the specified models and tested in the context of a use case. This allows us to benchmark the performance of the protocol with previously proposed strategies. Furthermore, the CPN model allows us to formally verify the protocol correctness.

3. Results

Our taxonomy of existing strategies for hedging liquidity position depreciation distinguishes three major categories: portfolio optimization approaches [5], derivative-based hedging methods [4], and data-driven liquidity-management techniques [3]. The literature provides useful mitigation mechanisms, but it shows that most solutions either depend on external derivatives markets or are based on portfolio management approaches.

The Goal model specifies how the objectives of the actors are met by the protocol. In particular, liquidity providers seek downside protection for their CLMM positions and risk takers seek income from the tokenized premium transfered to them in exchange for absorbing exposure. By following the Trusted DApp Modeling approach, we define the architecture of a system that includes smart contracts for position registration, tokenized position coverage issuance, collateral locking, premium accounting, and payout execution. The pricing mechanism modeled uses key variables such as volatility, expected utilization and time horizon to determine premiums. In this way, the protocol adapts premium price dynamically, based on the main determinants of LP risk.

Compared to other solutions, the proposed protocol enables risk averse users to achieve better performance in hedging position equity. Furthermore, it operates independently of the existing derivative markets, which may not be available for all token pairs.

References

[1] V. Mohan, Automated market makers and decentralized exchanges: a DeFi primer, Financial Innovation 8 (2022) 20. URL: https://doi.org/10.1186/s40854-021-00314-5. doi:10.1186/s40854-021-00314-5.

[2] J. Hasbrouck, T. J. Rivera, F. Saleh, An economic model of a decentralized exchange with concentrated liquidity, Management Science (2025). URL: https://doi.org/10.1287/mnsc.2024.04510. doi:10.1287/mnsc.2024.04510.

[3] H. Zhang, X. Chen, L. F. Yang, Adaptive liquidity provision in uniswap V3 with deep reinforcement learning, arXiv preprint arXiv:2309.10129 (2023). URL: https://doi.org/10.48550/arXiv.2309.10129. doi:10.48550/arXiv.2309.10129.

[4] A. Khakhar, X. Chen, Delta hedging liquidity positions on automated market makers, arXiv preprint arXiv:2208.03318 (2022). URL: https://doi.org/10.48550/arXiv.2208.03318. doi:10.48550/arXiv.2208.03318.

[5] D. Kroo, V. Gorgadze, G. Ovchinnikov, A. Barger, Optimal portfolios of liquidity positions, in: 2024 6th International Conference on Blockchain Computing and Applications (BCCA), 2024, pp. 774–781. URL: https://doi.org/10.1109/BCCA62388.2024.10844460. doi:10.1109/BCCA62388.2024.10844460.

[6] A. R. Hevner, S. T. March, J. Park, S. Ram, Design science in information systems research, MIS Quarterly 28 (2004) 75–105. URL: https://doi.org/10.2307/25148625. doi:10.2307/25148625.

[7] J.-h. Xu, K. Paruch, S. Cousaert, Y. Feng, SoK: Decentralized exchanges (DEX) with automated market maker (AMM) protocols, ACM Computing Surveys 55 (2023) 1–50. URL: https://doi.org/10.1145/3570639. doi:10.1145/3570639.

[8] C. J. Udokwu, A modelling approach for building blockchain applications that enables trustable inter-organizational collaborations, Doctoral dissertation, Lappeenranta-Lahti University of Technology LUT, 2022. URL: https://urn.fi/URN:ISBN:978-952-335-849-2.

[9] K. Jensen, L. M. Kristensen, L. Wells, Coloured petri nets and cpn tools for modelling and validation of concurrent systems, International Journal on Software Tools for Technology Transfer 9 (2007) 213–254.